THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 23, 4, 25, 6, 3, 19, 20 respectively, The mean of the data is: 1

(a) 270 **(b)** 269.8 **(c)** 270.8 **(d)** 268.8

2. A bag contains 15 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 3 **(b)** 44 **(c)** 60 **(d)** 45

3. The smallest positive integer by which 12 must be multiplied to obtain a perfect square is: 1

(a) 2 **(b)** 6 **(c)** 12 **(d)** 3

4. If α, β are zeroes of $x^2 - 6x + 8$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) 20 **(b)** $\frac{9}{2}$ **(c)** $\frac{5}{2}$ **(d)** $\frac{3}{4}$

5. The area (in sq. units) of the triangle formed by the line $2x + 3y = 12$ and the coordinate axes is: 1

(a) 72 **(b)** 12 **(c)** 3 **(d)** 24

6. If $x = 1$ is a root of $kx^2 + 3x - 5 = 0$, then $k =$ 1

(a) 2 **(b)** 5 **(c)** -2 **(d)** -8

7. The sum of the first n terms of an AP is $S_n = 2n^2 + 3n$. Its common difference is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 2

8. If $\triangle ABC \sim \triangle PQR$, $ar(ABC) = 18$ and $ar(PQR) = 98$ and $AB = 6$, then $PQ =$ 1

(a) 6 **(b)** 17 **(c)** 98 **(d)** 14

9. The ratio in which $(-3, 2)$ and $(6, 5)$ are joined by the y-axis is 3: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 9

10. If $\sec\theta + \tan\theta = 3$, then $\sec\theta - \tan\theta =$ 1

(a) 1 **(b)** $\frac{1}{3}$ **(c)** 3 **(d)** 2

11. From the top of a tower $6\sqrt{3}$ m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 30° and 60°. The distance between the two objects is: 1

(a) 6 m **(b)** 12 m **(c)** $12\sqrt{3}$ m **(d)** 24 m

12. Two concentric circles have radii 68 cm and 32 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 60 cm **(b)** 120 cm **(c)** 72 cm **(d)** 100 cm

13. A sector of a circle of radius 42 cm has central angle 45°. The perimeter of the sector is: 1

(a) 75 cm **(b)** 117 cm **(c)** 159 cm **(d)** 33 cm

14. A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

(a) 7 cm **(b)** 10.5 cm **(c)** 21 cm **(d)** 14 cm

15. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 120 **(b)** 100 **(c)** 103.6 **(d)** 108

16. A bag contains 7 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 4 **(b)** 6 **(c)** 10 **(d)** 5

17. **Assertion (A):** When an observer walks towards the foot of a tower, the angle of elevation of its top increases. 1

Reason (R): For a tower of fixed height h at horizontal distance d, $\tan\theta = \dfrac{h}{d}$, which increases as d decreases.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** A tangent PT touches a circle with centre O at P. If $OP = 5$ cm and $PT = 12$ cm, then $OT = 13$ cm. 1

Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** The area of a sector of a circle of radius 7 cm and angle 90° is $38.5\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The area of a sector of angle θ is $\dfrac{\theta}{360^\circ} \times \pi r^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The total surface area of a solid hemisphere of radius r is $3\pi r^2$. 1

Reason (R): The total surface area of a solid hemisphere is its curved surface area $2\pi r^2$ plus the area πr^2 of its flat circular face.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The mean of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 3, 24, p, 10, 15, 14, 17 respectively is 274. Find the value of p. 2

22. Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that the sum is a multiple of 4. 2

23. Find the smallest positive integer that leaves remainders 7 and 13 when divided by 12 and 18 respectively. 2

24. If the sum of the zeroes of $x^2 - 6x + k$ is equal to their product, find k. 2

25. Solve: $5x + 3y = 47$ and $3x + 5y = 41$. 2

26. If α, β are the roots of $x^2 - 6x + 5 = 0$, find $\alpha^2 + \beta^2$. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Find a and b for which the pair $3x + y = 7$ and $(a + b)x + (a - b)y = 14$ has infinitely many solutions. 3

28. For what values of k does $x^2 + 5x + k = 0$ have real roots? Find the greatest integer value of k. 3

29. Find the sum of the first 16 terms of the AP whose 3rd term is 14 and 8th term is 34. 3

30. Find the altitude of an equilateral triangle of side 6 cm, in terms of $\sqrt{3}$. 3

31. The line joining $(0,\ 0)$ and $(9,\ 6)$ is trisected by points P and Q. Find their coordinates. 3

OR

If $\sin\theta + \cos\theta = \frac{7}{5}$, find $\sin\theta\cos\theta$. 3

32. A flagstaff stands on the top of a building 14 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 36$ cm and $BC = 77$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Mohan starts a monthly savings plan. He saves ₹ 100 in the first month and increases his saving by ₹ 20 every month. **5**

(i) Write the first term and the common difference of the AP formed by his monthly savings.

(ii) How much does he save in the 10th month?

(iii) Find his total savings in the first 12 months.

(iv) In which month will he save ₹ 400, and what will his total savings be up to that month?

34. **Case study.** At the same time of the day, a girl of height 1.5 m casts a shadow 2 m long, while a tower casts a shadow 24 m long on level ground. **5**

(i) Name the similarity criterion that makes the two triangles formed (object, shadow, sun's ray) similar.

(ii) Find the height of the tower.

(iii) At that time, a pole 6 m high will cast a shadow of what length?

(iv) Find the distance from the top of the tower to the tip of its shadow.

35. Three vertices of a parallelogram $ABCD$ taken in order are $A(-2,\ 1)$, $B(1,\ 0)$ and $C(4,\ 3)$. Find the fourth vertex D. **5**

36. If $\tan A = \dfrac{24}{7}$, evaluate $\dfrac{5\sin A + 4\cos A}{7\sin A + 3\cos A}$. **5**

OR

Two vertical poles of equal height stand on opposite sides of a road 48 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). **5**

37. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 72 cm. The length of each tangent is 21 cm. Find the length of the chord AB. **5**

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 36° is marked in both. Find the area of the region between the two arcs within the sector. **5**

38. A solid sphere of radius 31.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). Find the volume of the space in the cylinder not occupied by the sphere. **5**

OR

The marks obtained by students in a test are given below. **5**

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	7	23	19	2	9	16	13	11

Find the mean and the mode (in this order).

END OF QUESTION PAPER — **Total: 80 Marks**